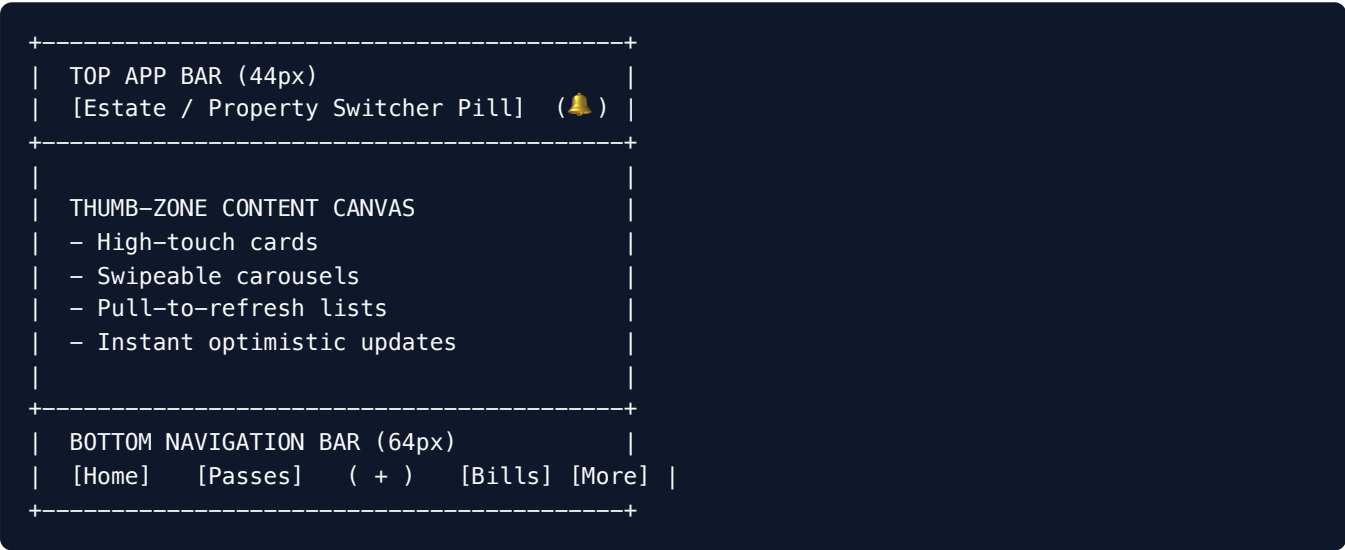


Estate Manager Web App — Mobile-First Page Architecture & Screen Inventory

Platform: Progressive Web Application (PWA)
UX Philosophy: Mobile-First Consumer Delight (Linear / Revolut caliber) + Ultra-Fast Gate Ergonomics
Target Viewport: 95%+ Mobile (iOS Safari / Android Chrome PWA) for Residents & Guards; Responsive Hybrid for Admins

1. Mobile-First UX Framework & Design Conventions

To displace 14-year legacy systems, the mobile interface must feel **indistinguishable from a native iOS/Android app**:



Core Mobile Conventions

- Persistent Thumb-Zone Bottom Navigation:** Primary resident routes accessible within one thumb tap.
- Bottom Sheet Action Modals:** Filtering, bill payment, and code creation slide up from the bottom rather than using desktop-style centered popups.
- PWA Standalone Display:** Runs without browser address bars (`"display": "standalone"` in `manifest.json`), with native iOS notch and Android safe-area insets.
- Haptic & Visual Feedback:** Tactile confirmation on code generation, bill payment, and guard gate approvals.
- Offline Fallback Engine:** PWA Service Worker serves cached active passes and vital estate contacts even in zero-connectivity elevator shafts or basements.

```

graph LR
    EMP[Estate Manager Pages] --- AO[Auth & Onboarding]
    EMP --- SA[Super Admin & Auditor]
    EMP --- EAP[Estate Admin Portal]
    EMP --- GGS[Guard Gate Station]
    EMP --- RPWA[Resident PWA]
    EMP --- OD[Operations Dashboard]

    AO --- AI[Accept Invitation]
    AO --- OF[Offline Fallback]
    AO --- L[Login]
    AO --- FP[Forgot Password]

    SA --- DCRH[Debt Code Restriction Hub]
    SA --- MEM[Multi-Estate Management]
    SA --- SAExpl[System Audit Explorer]
    SA --- SGA[Staff & Gate Allocations]

    EAP --- PR[Property Registry]
    EAP --- CK[Complaints Kanban]
    EAP --- FS[Facility Scheduler]
    EAP --- BSI[Bulk & Single Invoicing]
    EAP --- RD[Resident Directory]
    EAP --- GLM[Gate Live Monitor]

    GGS --- PL[PIN Login]
    GGS --- GCK[Gate Console Keypad]
    GGS --- ICM[Instant Camera Modal]
    GGS --- GSA[Gate SOS Alarm]
    GGS --- VH[Verification History]

    RPWA --- NPS[New Pass Sheet]
    RPWA --- NC[Notification Center]
    RPWA --- EN[Estate Noticeboard]
    RPWA --- MC[Mobile Checkout]
    RPWA --- NTS[New Ticket Sheet]
    RPWA --- ACH[Access Code Hub]
    RPWA --- BP[Bills & Payments]
    RPWA --- ESOS[Emergency SOS]
    RPWA --- FB[Facility Booking]
    RPWA --- PD[Profile & Dependents]
    RPWA --- CFM[Complaints & FM Chat]
    RPWA --- MP[My Properties]
    RPWA --- DF[Dashboard Feed]
    RPWA --- CF[Community Feed]

    OD --- ODash[Operations Dashboard]
    OD --- FSched[Facility Scheduler]
  
```

Section A: Authentication & Onboarding (Mobile Responsive)		
Route	Screen Name	Key Mobile UX / Micro-Interactions
/login	Unified Mobile Sign-In	Phone/Email input with biometric FaceID/TouchID (WebAuthn) prompt for returning users. Clean one-tap login.
/forgot-password	Password Recovery	4-digit SMS/Email OTP verification with automatic clipboard paste detection.
/accept-invite	Onboarding Activation	Welcomes residents synced from the external onboarding platform. Pre-fills property details, prompts to set password and install PWA.
/offline	PWA Offline Standby	Displays locally cached active access codes and offline estate emergency numbers when network is lost.

Standard Navigation: Fixed bottom tab bar with 4 tabs + 1 central floating action button (FAB).

- **Header Pill:** Multi-property dropdown switcher (e.g., Court A – A223 (Mansfield) ▼). Tapping switches context instantly across all data widgets.
- **Outstanding Dues Card:** Displays outstanding balance in high-contrast card. If debt exists, features a pulsing "Pay Now" button.
- **Active Guest Passes Carousel:** Swipeable cards showing currently active visitor codes with live countdown timers.

- **Quick Action Grid:** 1-tap shortcuts: *Book Amenity, Report Maintenance, Estate Notices, SOS.*
- **Recent Activity Feed:** Real-time log of recent gate entries, payment receipts, and announcement snippets.

2. Tab 2 — Access Code Center (`/resident/codes`)

- **Segmented Tabs:** `Active Passes (3)` | `History` / `Expired` .
- **Pass Card Details:** Visitor Name, Type badge (*Guest, Artisan, Delivery, Cab*), Expiry time, and single/multi-use tag.
- **1-Tap Share Button:** Opens native mobile share sheet (WhatsApp, SMS, Telegram) with pre-formatted invitation text and gate link.
- **Full-Screen QR Modal:** Tapping a pass enlarges a high-contrast QR code with automatic screen brightness boost for easy gate scanning.
- **Code Management:** 1-tap "Deactivate Pass" or "Extend Expiry by 2 Hours".

3. Center FAB — Quick Pass Generator (`/resident/codes/new`)

- *Renders as an ergonomic bottom sheet.*
- **Preset Selector:** One-touch pill selector (*Visitor, Cab / Uber, Artisan / Worker, Delivery*).
- **Validity Duration Picker:** Quick chips (+2 hrs, Today only, Custom date/time).
- **Restriction Warning (Rules 11–13):** If Super Admin disabled code generation due to arrears, the generator is replaced with a polite, non-punitive notice: *"Access code generation is temporarily paused. Pay outstanding service charges to restore immediately."* with a direct link to checkout.

4. Tab 3 — Bills & Payments (`/resident/bills`)

- **Financial Summary Banner:** Total Outstanding vs Total Paid this year.
- **Invoice List:** Color-coded status chips (`PENDING` , `PARTIALLY_PAID` , `PAID` , `OVERDUE`).
- **Tap-to-Expand Details:** Itemized breakdown (Service Charge, Security Levy, Water, Diesel).
- **Download Receipt:** 1-tap PDF invoice/receipt download to mobile storage.

5. Payment Checkout Sheet (`/resident/bills/:id/pay`)

- **Payment Selector:** Full payment vs Custom partial amount.
- **Gateway Embed:** Seamless modal supporting Apple Pay, Google Pay, Debit Cards, and USSD/Bank Transfer.
- **Instant Verification Screen:** Polished checkmark animation upon webhook confirmation, automatically refreshing the dashboard.

6. Tab 4 — More / Menu (`/resident/menu`)

Sub-pages accessible via the secondary navigation:

- **Complaints & Maintenance Hub (`/resident/complaints`):**
 - Ticket list with 5-stage progress indicator (`OPEN` $\rightarrow$ `ASSIGNED` $\rightarrow$ `IN TREATMENT` $\rightarrow$ `RESOLVED` $\rightarrow$ `CLOSED`).
 - FAB: `/resident/complaints/new` (Bottom sheet with camera capture for leaky pipes, electrical faults, noise).
 - Ticket Thread: `/resident/complaints/:id` (Two-way WhatsApp-style live chat with assigned FM staff).
- **Facility Booking (`/resident/facilities`):**

- Photo gallery of estate amenities (Swimming Pool, Tennis Court, Gym, Event Hall).
- Interactive mobile time-slot grid with conflict-prevention (booked slots disabled in grey).
- **Estate Noticeboard & Messages (/resident/messages):**
 - Official estate broadcast announcements with push-notification deep links.
- **Community Feed (/resident/community):**
 - Estate-scoped neighborhood discussion board with photo sharing and resident comments.
- **Emergency SOS Hub (/resident/sos):**
 - Prominent red guarded button (requires 3-second hold to prevent accidental triggers).
 - Instant real-time siren broadcast to gate guards and FM team with property location.
 - Quick-call directory for local police, ambulance, and fire services.
- **My Properties & Dependants (/resident/properties):**
 - Overview of all assigned units across estates, including registered dependants, household staff, and vehicles.
- **Notification Center (/resident/notifications):**
 - Chronological stream of gate entry verifications, payment receipts, and announcements.

Section C: Security Guard Gate Station (Handheld / Tablet Optimized)

Designed for high-glare sunlight, night shifts, and single-hand or glove use at estate gates.

+-----+-----+			
	GATE STATION – GATE 1 (NORTH)		
+-----+-----+			
	[ENTER 6-DIGIT CODE]		
	[1]	[2]	[3]
	[4]	[5]	[6]
	[7]	[8]	[9]
	[SCAN QR]	[0]	[CLEAR]
+-----+-----+			
	VERIFY CODE -> AUTO CAMERA VIEWFINDER		
+-----+-----+			

Route	Screen Name	Key Mobile UX / Micro-Interactions
/guard/gate-station	The Verification Console	The Hero Gate Screen. Huge numeric keypad + QR scanner toggle. Sub-5-second workflow: 6 digits entered $\rightarrow$ instant validation card appears $\rightarrow$ camera opens $\rightarrow$ snap photo $\rightarrow$ tap "Approve".
/guard/camera-modal	Gate Photo Capture	Direct HTML5 camera integration. Auto-prompts for Visitor Photo (mandatory) and Vehicle Plate Photo (if flagged). Retake / Confirm toggles. Background upload.
/guard/history	Today's Gate Entries	Real-time scrollable timeline of approved/rejected entries with visitor thumbnail photos, timestamps, and destination property.
/guard/sos-alerts	Emergency Gate Alert	Full-screen flashing alarm overlay triggering audio alert when a resident initiates an SOS, showing exact court, block, and apartment.

Section D: Estate Admin & Facility Management (Responsive Web & Mobile)

Optimized for FM managers moving around the estate on mobile devices or managing from office desktops.

Route	Screen Name	Key Features & Layout
/admin/dashboard	Operations Command Center	Real-time KPI summary widgets: Total Residents, Today's Gate Entries, Unpaid Invoices, Open Tickets, Active SOS Alerts.
/admin/residents	Resident Directory	Filterable list/table by court/block. One-tap phone dialer, unit mapping, and expandable modal for authorized dependants.
/admin/properties	Property Registry	Hierarchical browser: Estate $\rightarrow$ Court $\rightarrow$ Block $\rightarrow$ Unit with occupancy state (Owner-occupied, Tenant, Vacant).
/admin/billing	Invoicing & Arrears Hub	Bulk billing wizard (distribute service charges to all 200 units in 3 steps); individual invoice generator; arrears aging report.
/admin/complaints	Maintenance Kanban Board	Drag-and-drop ticket columns (Open , Assigned , In Treatment , Resolved). One-click staff assignment.
/admin/broadcasts	Announcement Broadcaster	Multi-channel broadcast tool (In-app notice, Web Push, SMS) segmented by estate or specific court.
/admin/gate-feed	Live Gate Surveillance	Real-time streaming feed of all gate verifications, guard approvals, and visitor image captures.
/admin/facilities	Amenity Management	Operating hours configuration, maintenance blockout dates, and resident booking approval queue.

Section E: Super Admin & Auditor (Executive Governance)

Route	Screen Name	Key Features & Layout
/super-admin/estates	Global Estate Portfolio	Multi-estate configuration: estate branding, gate photo rules (toggle mandatory visitor/vehicle image), address.
/super-admin/staff	Staff & Permissions	Provisioning Admins and Guards; assigning admins to specific estate scopes (EstateAdminAssignment).
/super-admin/access-restrictions	Debt Restriction Console (Rules 11–13)	Search delinquent accounts; one-click "Restrict Code Generation" with mandatory reason dropdown and audit note; one-click restore.
/super-admin/audit-logs	Global Audit Explorer	Immutable system-wide log explorer: filter by actor, estate, action type, IP address, and date range.
/auditor/overview	Auditor Compliance Portal	Read-only financial ledgers, gate entry archives, and compliance export tool (CSV/PDF).

4. Route Manifest Summary Table

```
/ (Redirects based on authenticated role)
├── /login
├── /forgot-password
├── /accept-invite
├── /offline
├── /resident (Role: RESIDENT)
│   ├── /dashboard
│   ├── /properties
│   ├── /codes
│   │   └── /new (Bottom Sheet)
│   ├── /bills
│   │   └── /:id/pay
│   ├── /complaints
│   │   ├── /new
│   │   └── /:id (Chat)
│   ├── /facilities
│   │   └── /my-bookings
│   ├── /messages
│   ├── /community
│   ├── /sos
│   ├── /notifications
│   └── /profile
├── /guard (Role: GUARD)
│   ├── /gate-station
│   ├── /history
│   └── /sos-alerts
├── /admin (Role: ADMIN)
│   ├── /dashboard
│   ├── /residents
│   ├── /properties
│   ├── /billing
│   │   └── /bulk
│   ├── /complaints
│   ├── /broadcasts
│   ├── /gate-feed
│   └── /facilities
├── /super-admin (Role: SUPER_ADMIN)
│   ├── /estates
│   ├── /staff
│   ├── /access-restrictions
│   └── /audit-logs
└── /auditor (Role: AUDITOR)
    └── /overview
```